Name Surname
Class: XE

Laboratory of xxxx

Report Template

Name Surname

October 28, 2026

Abstract

The technical report that concludes a laboratory experiment communicates the aims of the work, how it was carried out and the results obtained.

It must be written so that anyone can reproduce the experiment and compare the results.

For this reason the report is organised, in order, into the sections that follow. Each section is described below, with suggestions on how to write it. Replace this text with a short summary of your experiment, its method and its main result.

1 Objective

Sometimes called *Aim* or *Purpose*.

State the objectives of the experiment in one or two sentences. This section can be omitted when the aim is already clear from the title.

Example: "Purification of benzoic acid contaminated with activated carbon by recrystallisation."

2 Principle of the method

Sometimes called *Theory* or *Theoretical background*.

Describe the theoretical principles the experiment is based on and how they are used to reach the objectives. Include every concept needed to understand the report. If the report belongs to a series, you may refer to earlier reports instead of repeating the same theory.

A glossary-style list can work well:

Stoichiometry The relationship between the relative quantities of substances taking part in a reaction or forming a compound, typically a ratio of whole numbers.

Atomic mass The mass of an atom of a chemical element expressed in unified atomic mass units (Da). It is approximately equal to the number of protons and neutrons in the atom (the mass number) or, for an element, to the average weighted by the natural abundance of its isotopes.

Alternatively, use `\subsection{...}` to organise longer explanations, and number the key equations so that you can refer to them later with `\cref{eq:label}`.

3 Instruments

Sometimes called *Equipment* or *Apparatus*.

List the glassware and instrumentation used, with the technical data that matter for the result: capacity, tolerance (or class) and sensitivity of the glassware, model and resolution of the instruments.

Example: burette, capacity 25 mL, tolerance ±0.05 mL.

A bulleted list, nested if needed, groups the items clearly:

- Instruments:
  - microscope slide;
  - Bunsen burner;
  - inoculating loop;
  - wooden tongs;
  - burette, capacity 25 mL, tolerance ±0.05 mL;
  - …
- Culture media:
  - malt agar;
  - nutrient agar.
- Dyes:
  - methylene blue;
  - gentian violet.

When the list is long, two columns save space:

- Instruments:
  - microscope slide;
  - Bunsen burner;
  - inoculating loop;
  - wooden tongs;
  - …
- Culture media:
  - malt agar;

  - nutrient agar.
  - methylene blue;
- Dyes:
  - gentian violet.

A table is another option (see Table 1).

Instruments	**Culture media**	**Dyes**
Microscope slide	Malt agar	Methylene blue
Bunsen burner	Nutrient agar	Gentian violet
Inoculating loop		
Wooden tongs		
...		

Table 1: Instruments and materials used in the experiment.

4 Reagents

Sometimes called *Materials and chemicals*.

State the type and characteristics of the reagents used: always give the concentration of solutions and, for solids, the purity grade. When a safety data sheet (SDS) is available, summarise its key information: pictograms, H and P statements, personal protective equipment (PPE) and occupational exposure limits.

Chemical formulae are written with the `mhchem` package and quantities with `siunitx`:

- H_3PO_4, 0.3 mol/L, 25 mL;
- H_2O_2, 30 % (w/w), 10 mL.

Table 2 shows a compact safety table; the P statements are written as codes and can be listed in full at the end of the report.

Table 3 extends the previous table with the occupational exposure limits: concentrations of airborne substances below which most workers can be exposed repeatedly, day after day, without adverse health effects. The commands `\Hphrase` and `\Pphrase` print the full text of each statement.

For syntheses a different table is often more useful (Table 4): it collects stoichiometric ratios, masses and yields, which help to follow the mechanism when it depends on the reagent quantities.

To find the PPE, search the web for the compound name followed by "safety data sheet" and use the most recent result.

Example: "benzoic acid safety data sheet".

Note the manufacturer of each reagent (a photo of the bottle in the lab is enough): it speeds up the search and makes the data more accurate, since sheets from different suppliers may differ. Besides the PPE, the SDS contains much more useful information.

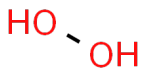
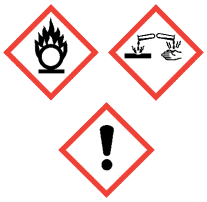
Compound	Structure	H and P statements	Pictograms	PPE
H_2O_2		H: H271, H302, H314, H332. P: P210, P220, P221, P260, P261, P264, P270, P271, P280, P283, P301+P312, P301+P330+P331, P303+P361+P353, P304+P312, P304+P340, P305+P351+P338, P306+P360, P310, P312, P321, P330, P363, P370+P378, P371+P380+P375, P405, P501.		gloves, goggles, face shield

Table 2: Chemicals used in the experiment. The H and P statements are given in full at the end of the report.

5 Procedure

Sometimes called *Method* or *Experimental*.

Report the operating procedure that was actually followed, concisely but completely, so that the experiment can be reproduced. A schematic drawing of the apparatus helps when the assembly is not obvious.

Comments and observations are welcome when they are relevant, for instance details not mentioned in the given procedure but noticed during the experiment.

Example: Add Devarda's alloy. *Warning: after the addition the test tube becomes very hot; work near the bench and keep a beaker at hand to rest the tube in.*

A flowchart that summarises the steps is also useful, both for clarity and for studying (see the flowchart example at the end of this document).

6 Reactions

Sometimes called *Reaction mechanism*.

If the experiment involves chemical reactions, write them down and explain them: this is one of the key parts of a chemistry report. Use the `\ce{...}` command of the `mhchem` package for formulae and equations, and the `equation` environment to number them.

$$2\,H_2O_2 \longrightarrow 2\,H_2O + O_2\uparrow \tag{1}$$

Some `mhchem` examples:

- ions and charges: SO_4^{2-}, $[Cu(NH_3)_4]^{2+}$;
- equilibrium: $H_2O \rightleftharpoons H^+ + OH^-$;
- bonds: $HC{\equiv}CH$, $H_2C{=}CH_2$, $H_3C{-}CH_3$;

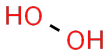
Compound	Structure	H and P statements	Pictograms	PPE	Limits [ppm]
H_2O_2		**H271** – May cause fire or explosion; strong oxidizer. **H302** – Harmful if swallowed. **H314** – Causes severe skin burns and eye damage. **H332** – Harmful if inhaled. **P280** – Wear protective gloves/protective clothing/eye protection/face protection/hearing protection/... **P305+P351+P338** – IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.		gloves, goggles, face shield	LTEL: 1 STEL: 2

Table 3: Chemicals used in the experiment with their exposure limits. LTEL: long-term exposure limit (8-hour time-weighted average); STEL: short-term exposure limit (15 minutes).

Substance	Molar mass [g/mol]	Amount [mol]	Stoich. ratio	Mass [g]	Volume [mL]	Yield [%]	H statements	P statements
A	B	C	D	E	F	G	H	I

Table 4: Table for syntheses.

- conditions above the arrow: $\mathrm{A} \xrightarrow{\Delta} \mathrm{B}$;
- states of matter and precipitates: $\mathrm{Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)\downarrow}$.

Refer to numbered reactions with `\cref`: the decomposition in Equation (1) is catalysed by MnO_2.

7 Data and calculations

Collect here all the results obtained: photos, data tables and personal observations are all welcome. When the data are numerous, gather them in tables. The experimental data and the data processing can also be merged into a single narrative if that reads better.

Write every formula used for the calculations, explaining what it is for and giving the units of each quantity. There are two common layouts: write all the formulae first and then the

Compound	Structure	Appearance	Hazards	Protection	TLV/TWA	Disposal
H_2O		colourless liquid	none	none	not relevant	down the sink
NaCl		white solid	none	none	not relevant	down the sink

Table 5: Another reagent table layout.

calculations with your data, or write each formula immediately followed by its calculation. A table suits the second layout well.

First layout. The amount of substance is obtained from the molar concentration c and the volume V of the solution:

$$n = c \cdot V \tag{2}$$

where n is in mol, c in mol/L and V in L. Applying Equation (2) to the data (the starred environment `equation*` suppresses the number):

$$n = 0.5\,\text{mol/L} \cdot 25 \times 10^{-3}\,\text{L} = 1.25 \times 10^{-2}\,\text{mol}$$

The volume is divided by 1000 because it was measured in millilitres.

Second layout.

Formula		**Calculation**
$n = c \cdot V$	$\rightarrow$	$n = 0.5\,\text{mol/L} \cdot 25 \times 10^{-3}\,\text{L} = 1.25 \times 10^{-2}\,\text{mol}$
$m = n \cdot M$	$\rightarrow$	$m = 1.25 \times 10^{-2}\,\text{mol} \cdot 34.01\,\text{g/mol} = 0.425\,\text{g}$
$c = m/V$	$\rightarrow$	$c = 0.425\,\text{g}/0.100\,\text{L} = 4.25\,\text{g/L}$

7.1 Experimental data

Report *all* the experimental data, highlighting where necessary the outliers, that is the values rejected on the basis of a statistical test. Whenever possible, collect the data in tables; the `S` column type of `siunitx` aligns numbers on the decimal point (Table 6).

7.2 Data processing

Report the calculations made from the experimental results, stating the mathematical relationships used. Processing may also include diagrams and plots, and sometimes the statistical treatment of the data. Plots are drawn with `pgfplots`, either from coordinates typed in the source (Figure 1) or from a data file (Figure 2).

Trial	Volume [mL]	Absorbance
1	10.05	0.512
2	15.10	0.768
3	20.00	1.021

Table 6: Experimental data.

8 Conclusions

Here the operator reports any notes on the procedure and evaluates the processed results against the objectives of the experiment. Point out any anomalies observed in the method and any step that caused difficulties, especially with regard to safety.

This is the last section of the report and, together with the objective and the procedure, perhaps the most important one. Draw the conclusions of the experiment:

- outcome (successful or not);
- result (plausible or not), comparing it with literature values or with values calculated from the literature;
- personal observations on steps that may have gone wrong, or that you believe you carried out better than the given instructions, with an explanation.

9 Bibliography

Not always required, but citing the sources used to write any document is good practice (the teacher's own material can usually be omitted). This section collects the websites, books and articles that provided ideas or information, especially for the theoretical background.

It also helps if you want to retrieve the sources later to study, or if what you wrote is questioned.

Add each source to `bibliography.bib` and cite it in the text with `\autocite{key}`.

Citation example.[1]

(1) Einstein, A. Zur Elektrodynamik bewegter Körper. *Annalen der Physik* **1905**, *322*, 891–921, DOI: 10.1002/andp.19053221004.

Hazard and precautionary statements

If you prefer not to write the full H and P statements in the reagent tables, list them here. Complete lists are available from PubChem and ECHA.

The template includes the GHS statements in `config/ghs_phrases.tex`, so the LaTeX source stays short: write only the code, as in `\Hphrase{H314}` or `\Pphrase{P305+P351+P338}`, and the full text is printed for you.

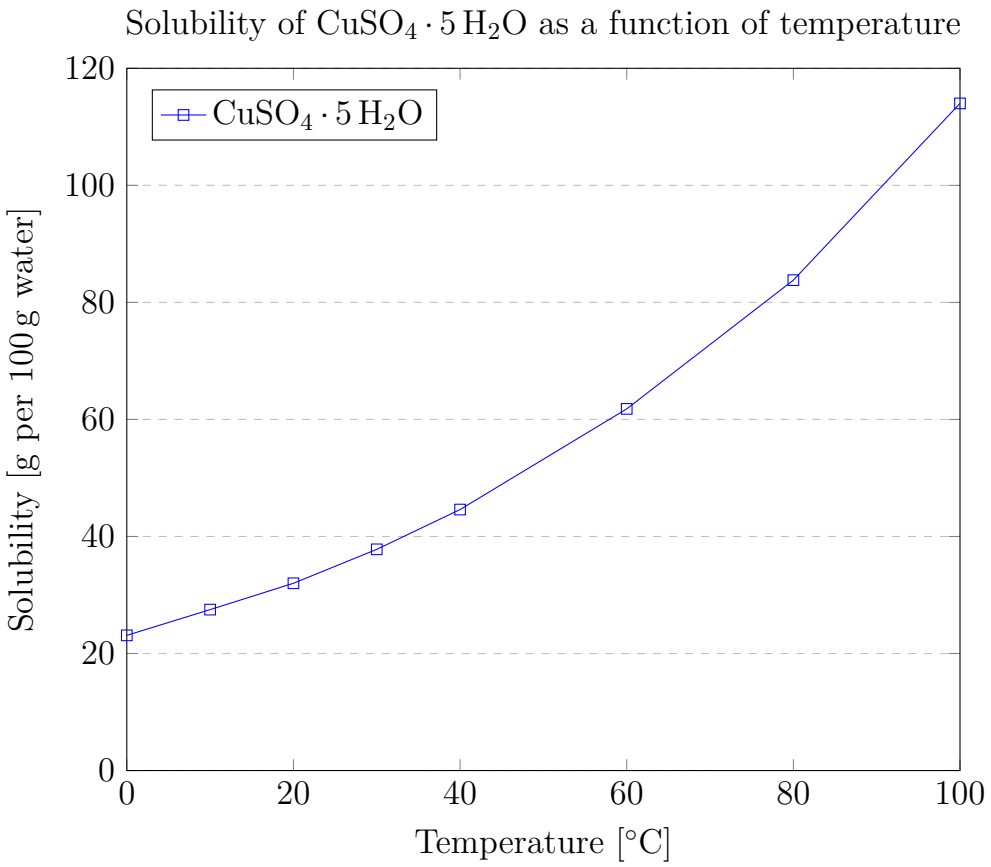


Figure 1: Solubility of copper(II) sulfate pentahydrate in water as a function of temperature.

Physical hazards

- **H271** – May cause fire or explosion; strong oxidizer.
- **H290** – May be corrosive to metals.

Health hazards

- **H302** – Harmful if swallowed.
- **H314** – Causes severe skin burns and eye damage.
- **H332** – Harmful if inhaled.

Precautionary statements

- **P210** – Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
- **P280** – Wear protective gloves/protective clothing/eye protection/face protection/hearing protection/...
- **P301+P330+P331** – IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- **P305+P351+P338** – IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- **P501** – Dispose of contents/container to ...

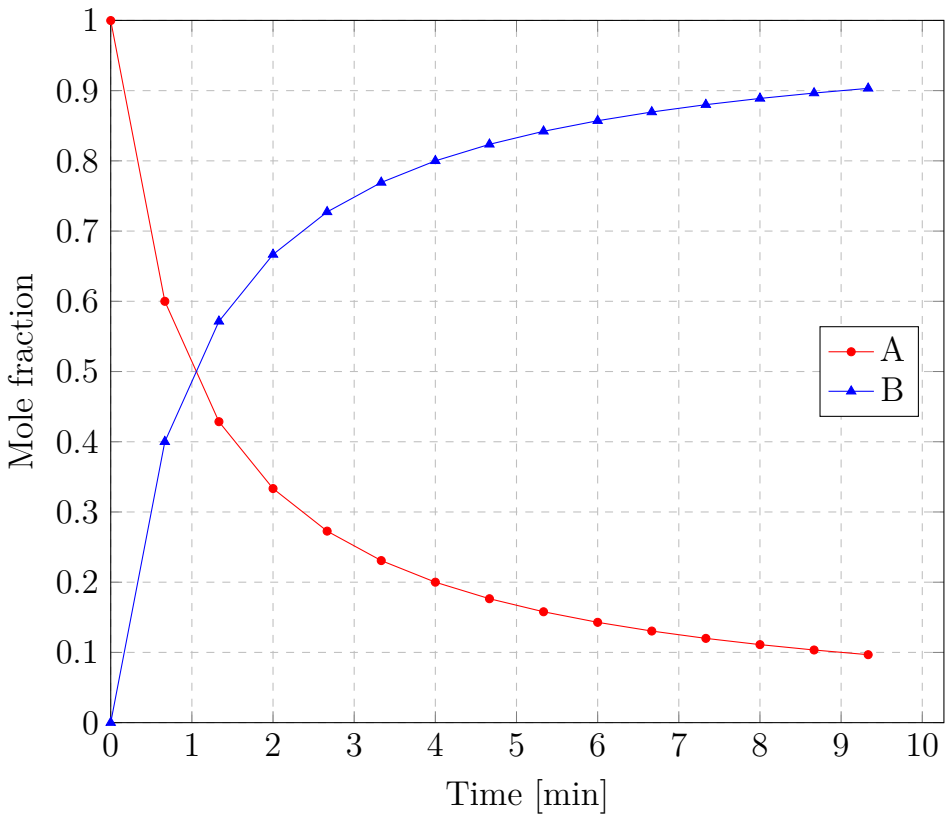


Figure 2: Reaction A $\longrightarrow$ B: mole fractions read from `data/data.txt` and `data/data2.txt`.

Useful links

A small collection of websites that help when writing a report.

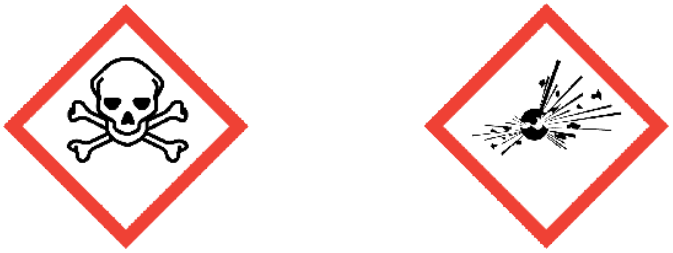
- MolView: 2D and 3D images of chemical compounds;
- Mol2chemfig: draws molecules and mechanisms and converts them into LaTeX code for the `chemfig` package;
- ECHA – CLP pictograms: the European hazard pictograms for the labelling of chemicals;
- ECHA – C&L Inventory: classification and labelling of substances in the EU;
- PubChem: the world's largest collection of freely accessible chemical information: properties, safety and toxicity data, literature and patents;
- ChemSpider: free chemical structure database with text and structure search over more than 100 million structures;
- CAS Common Chemistry: open resource with about 500 000 substances from the CAS REGISTRY, including the common ones used in school and undergraduate chemistry;
- ChEBI: Chemical Entities of Biological Interest, a dictionary of small molecular entities;
- ChEMBL: curated database of bioactive molecules with drug-like properties;
- DrugBank: drug interactions, pharmacology, chemical structures, targets and metabolism;
- RCSB PDB: Protein Data Bank, the public archive of 3D structures of proteins and nucleic acids;
- UniProt: protein sequences and functional information;
- enviPath: biodegradation pathways of chemicals (successor of the EAWAG-BBD);
- EFSA: European Food Safety Authority, scientific risk assessments and data;
- ARS: Agricultural Research Service of the US Department of Agriculture;
- NIST: National Institute of Standards and Technology; the NIST Chemistry WebBook provides thermochemical and spectroscopic data;
- The Merck Index Online: authoritative reference on chemicals, drugs and biologicals;
- WebElements: the periodic table and the properties of the elements;
- Molecule of the Month: one of the longest-running chemistry websites, with a new molecule each month since 1996;
- Chemistry Stack Exchange: questions and answers on chemistry;
- TeX Stack Exchange: questions and answers on LaTeX;
- mhchem, chemfig, siunitx, pgfplots: documentation of the main packages used by this template;
- annotate-equations: package to annotate the terms of an equation.

Examples

This section collects ready-to-copy examples of the most common elements of a report. Delete it, together with the useful links, in your final document.

Figures

Two images side by side, each with its own sub-caption (`subfigure` environment, Figure 3).



(a) Acute toxicity (b) Explosive

Figure 3: Two sub-figures centred on the page.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas eget nisl elementum, pretium libero suscipit, interdum tellus. Praesent luctus commodo massa, vel molestie augue laoreet ac. Phasellus sodales auctor erat eu porta. Donec at volutpat nunc. Phasellus pretium, eros vitae cursus ornare, turpis massa egestas sem, ut varius ipsum metus nec mi.

A figure with the caption beside it (`floatrow` package, Figure 4).

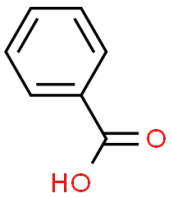


Figure 4: Benzoic acid: a figure with its caption on the right.

Nullam vel lorem porttitor, convallis ex eget, condimentum velit. Integer sed sem aliquet, elementum ipsum in, rutrum elit. Vestibulum in arcu eget odio egestas varius. Vivamus lacus augue, dignissim at arcu a, consequat iaculis leo. Nam sit amet varius tellus. Nulla tempor velit nibh, et tincidunt diam porta vel. Mauris sit amet erat ut neque vehicula ultrices et eu sem. Phasellus pellentesque ultricies sapien. Curabitur at sodales mauris, maximus auctor mi. Vestibulum semper mauris in euismod rutrum. Fusce pellentesque in mi ac euismod. Nam gravida magna ut volutpat placerat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Sed consequat justo condimentum metus bibendum mattis. Etiam id euismod ante. Nam sit amet ex libero.



Figure 5: A figure on the right, wrapped by the text.

The GHS hazard pictograms are available in `img/pictograms/` (Figure 7).

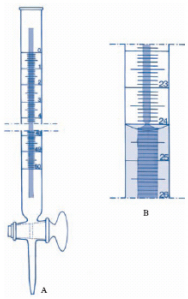


Figure 6: A burette.



Figure 7: GHS pictograms: file name and code.

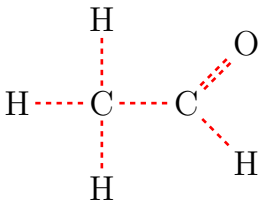
Units and cross-references

Quantities are typeset with `siunitx` and labelled objects are referenced with `cleveref`, which adds the word (Figure, Table, Equation) automatically.

Code	Result
`\qty{25.00}{\milli\litre}`	25.00 mL
`\qty{9.81}{\metre\per\second\squared}`	9.81 m/s^2
`\qty{0.1032(4)}{\mole\per\litre}`	(0.1032 ± 0.0004) mol/L
`\num{6.022e23}`	6.022×10^{23}
`\ang{104.5}`	104.5°
`\cref{fig:burette}`	Figure 6
`\cref{tab:reagents,tab:synthesis}`	Tables 2 and 4
`\cref{eq:moles}`	Equation (2)

Structural formulae

Structural formulae are drawn with the `chemfig` package (Table 7).
Molecules can be named and the bond style changed (Figure 8).



Ethanal

Figure 8: Named molecule with a custom bond style.

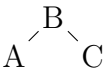
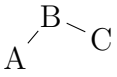
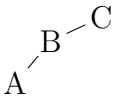
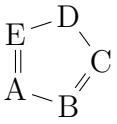
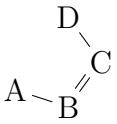
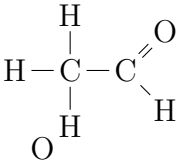
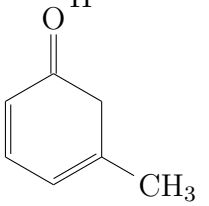
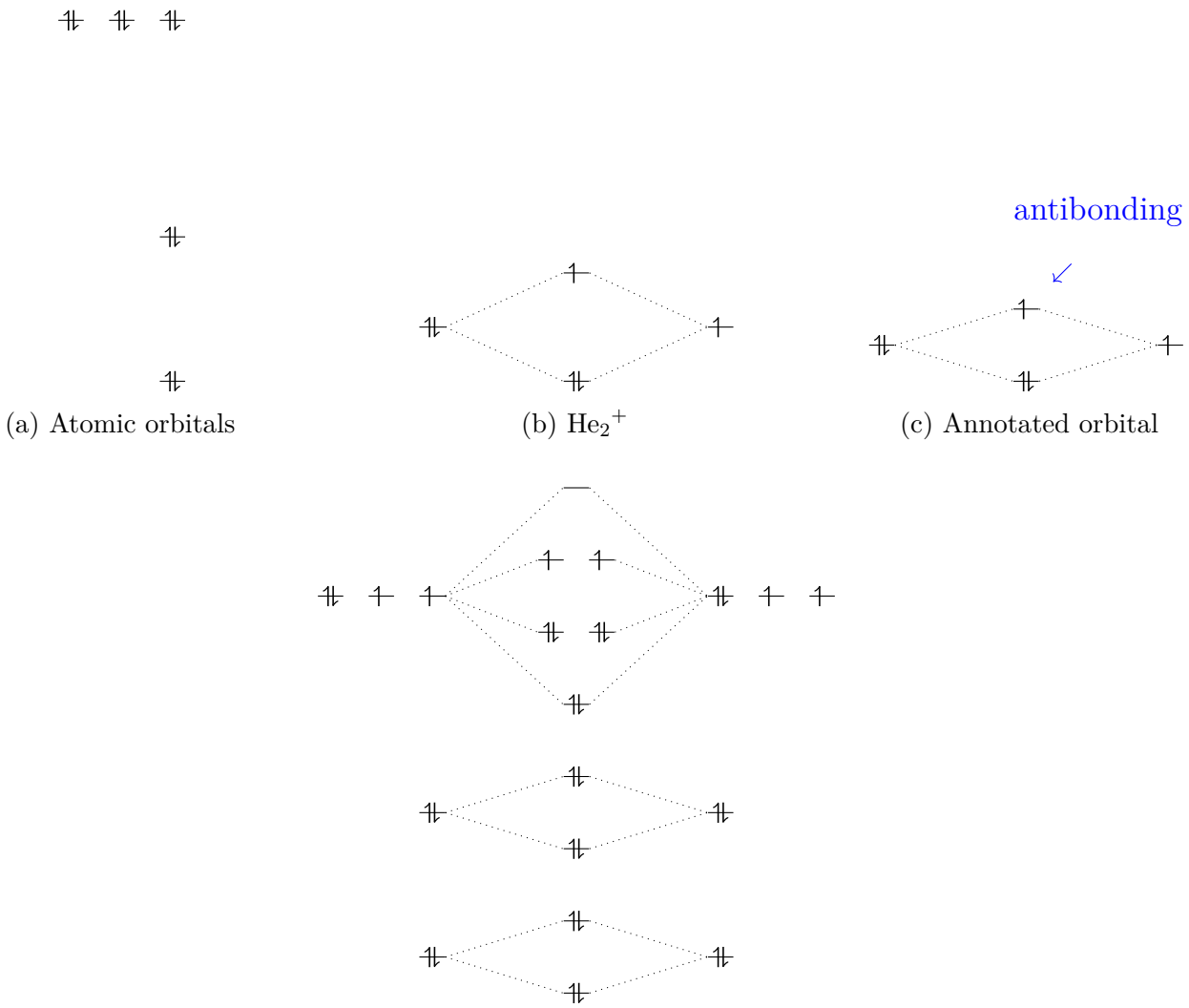
Feature	**Code**	**Result**
Bond angles in units of 45°	`\chemfig{A-[1]B-[7]C}`	
Absolute angles	`\chemfig{A-[:50]B-[:-25]C}`	
Relative angles	`\chemfig{A-[::50]B-[::-25]C}`	
Regular polygons	`\chemfig{A*5(-B=C-D-E=)}`	
Incomplete rings	`\chemfig{A*5(-B=C-D)}`	
Branched molecule	`\chemfig{H-C(-[2]H)(-[6]H)-C(=[1]O)-[7]H}`	
Branched ring	`\chemfig{*6(-=(-CH_3)--(=O)-=)}`	

Table 7: Some `chemfig` features.

Molecular orbital diagrams

Molecular orbital diagrams are drawn with the `modiagram` package (Figure 9).



(a) Atomic orbitals (b) He_2^+ (c) Annotated orbital

(d) O_2

Figure 9: Examples of molecular orbital diagrams.

Flowchart

A flowchart is useful for long or branched procedures (Figure 10). The node and arrow styles are defined in `config/flowchart.tex`.

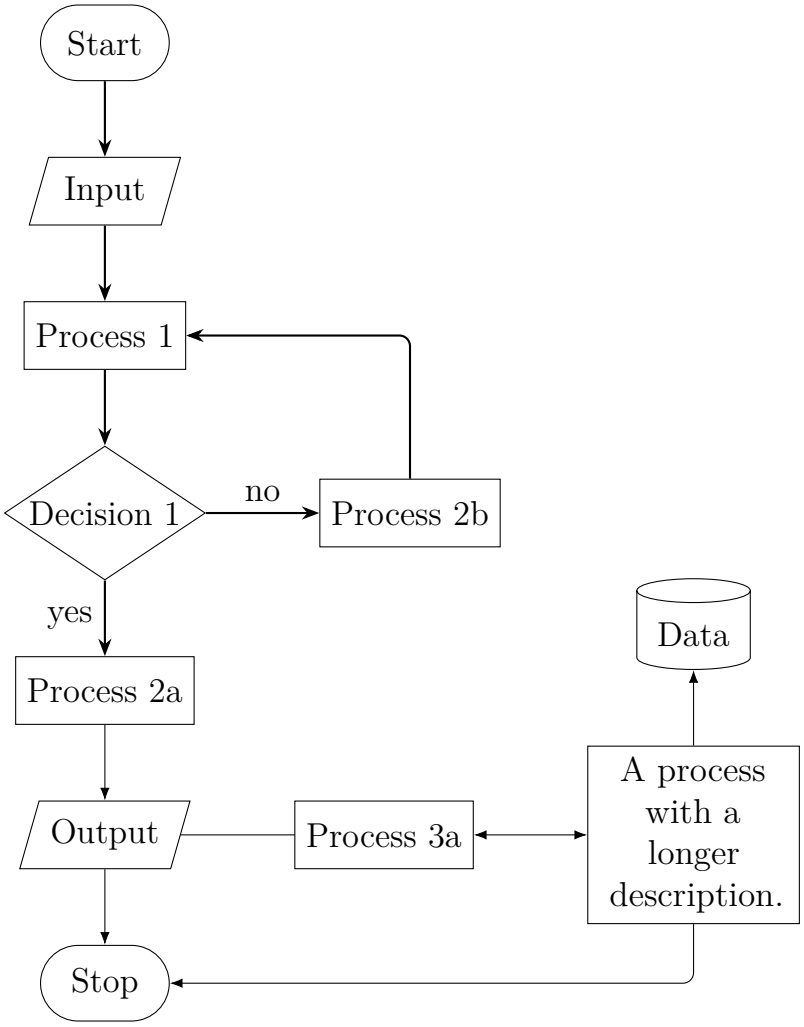


Figure 10: Example flowchart.

Notes

Notes are added with `\unsure`, `\change`, `\info` and `\improvement`, defined in `config/settings.tex`. The optional argument `[inline]` places a note in the text instead of the margin. To hide all notes, add the `disable` option to `todonotes` in `config/package.tex`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed eget velit ullamcorper, convallis neque eget, tempus leo. Aenean eget est ornare, mattis turpis sit amet, vestibulum ante. Integer quis molestie arcu. Duis sem felis, posuere ut ante vitae, lobortis tincidunt dolor. Ut aliquet nunc sed lorem vehicula, ac tristique velit venenatis.

Is this correct?

Change this!

Pellentesque consectetur malesuada lectus, ut faucibus diam egestas ac. Aenean porttitor at libero a venenatis. Morbi sollicitudin, leo sed pellentesque facilisis, lacus diam lobortis tellus, sit amet vulputate turpis sem id ipsum. Nunc ac aliquet mi, non porta quam. Pellentesque libero nunc, congue vitae mi sit amet, lobortis faucibus ante.

This can help me in the conclusions.

Sed non erat metus. Donec aliquet ex non neque sodales pretium. Aliquam eu sapien elit. Aliquam finibus felis et neque elementum, laoreet tempus urna auctor. Vivamus ac congue elit, vitae volutpat nulla.

This really needs to be improved!

The following paragraph needs to be rewritten!

Sed finibus pellentesque diam, et sagittis tortor ultrices ut. Suspendisse volutpat ullamcorper dui in cursus. Proin ullamcorper neque posuere porttitor rutrum.

Finally, `\listoftodos` prints the list of all notes:

Notes to address

Example: “Purification of benzoic acid contaminated with activated carbon by recrystallisation.” . . . 1
Example: burette, capacity 25 mL, tolerance ±0.05 mL. . . . 2
Example: “benzoic acid safety data sheet”. . . . 3
Example: Add Devarda’s alloy. *Warning: after the addition the test tube becomes very hot; work near the bench and keep a beaker at hand to rest the tube in.* . . . 4
Citation example.[1] . . . 7
Is this correct? . . . 17
Change this! . . . 17
This can help me in the conclusions. . . . 17
This really needs to be improved! . . . 17
The following paragraph needs to be rewritten! . . . 17